# MD-Informed Microfields of HED Plasmas for Line Shape Calculations

DSTI Exit Talk

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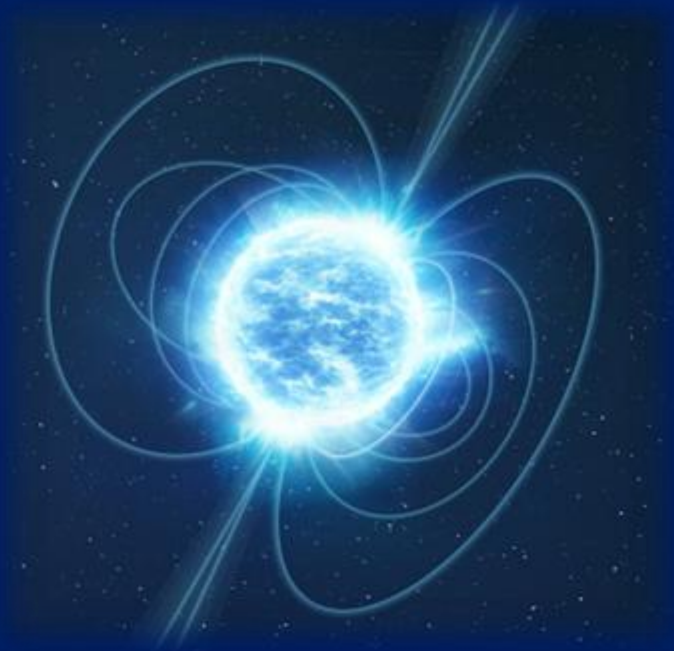
# Plasma

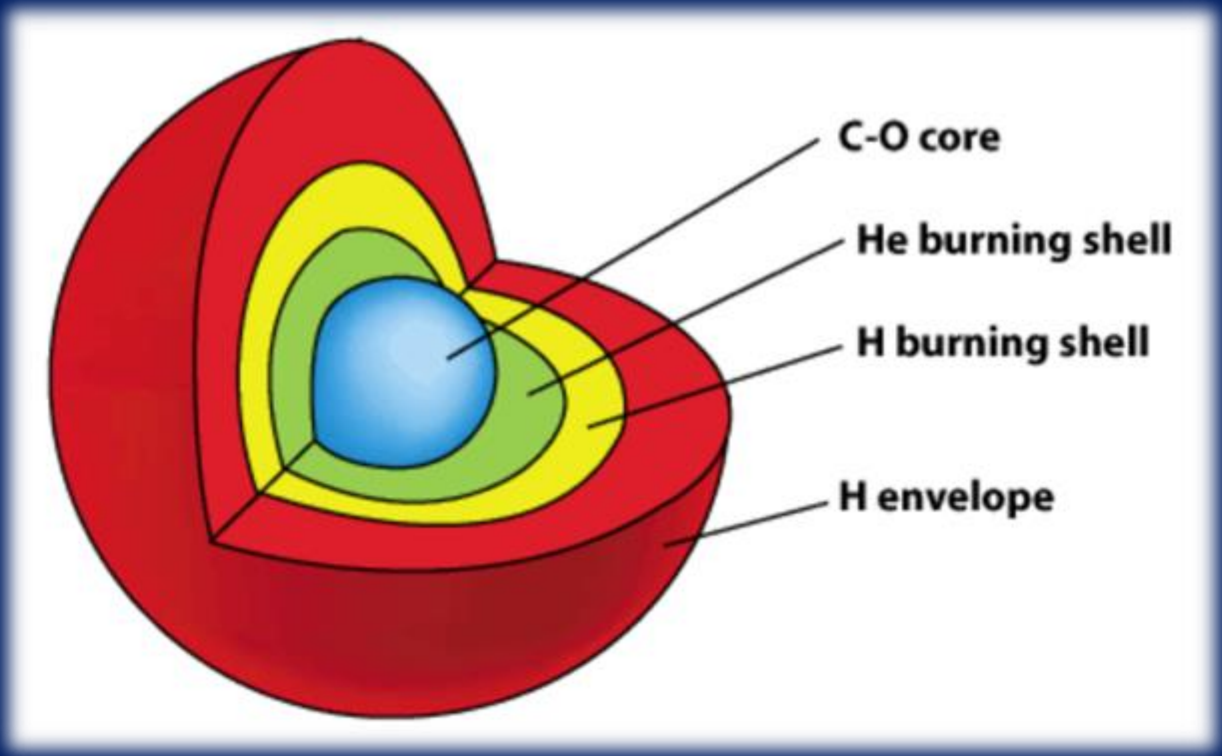
- Partially or fully ionized gas
- Found in many places
- Can behave as fluid or solid
- Stars and compact objects contain HED plasmas



ESA

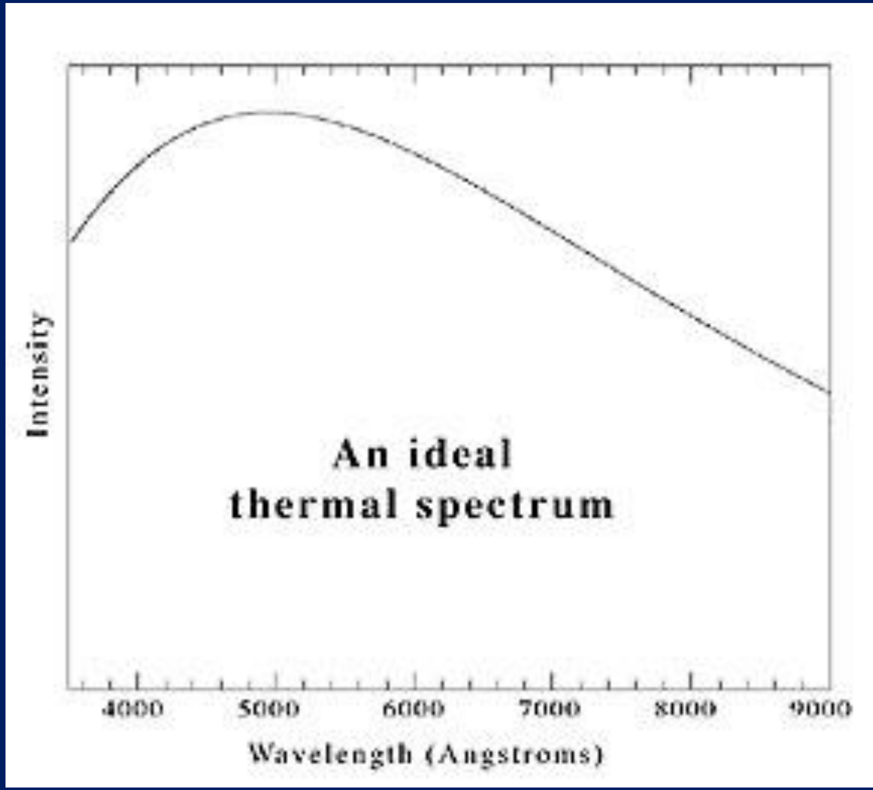
# Stellar spectra contain information about stars

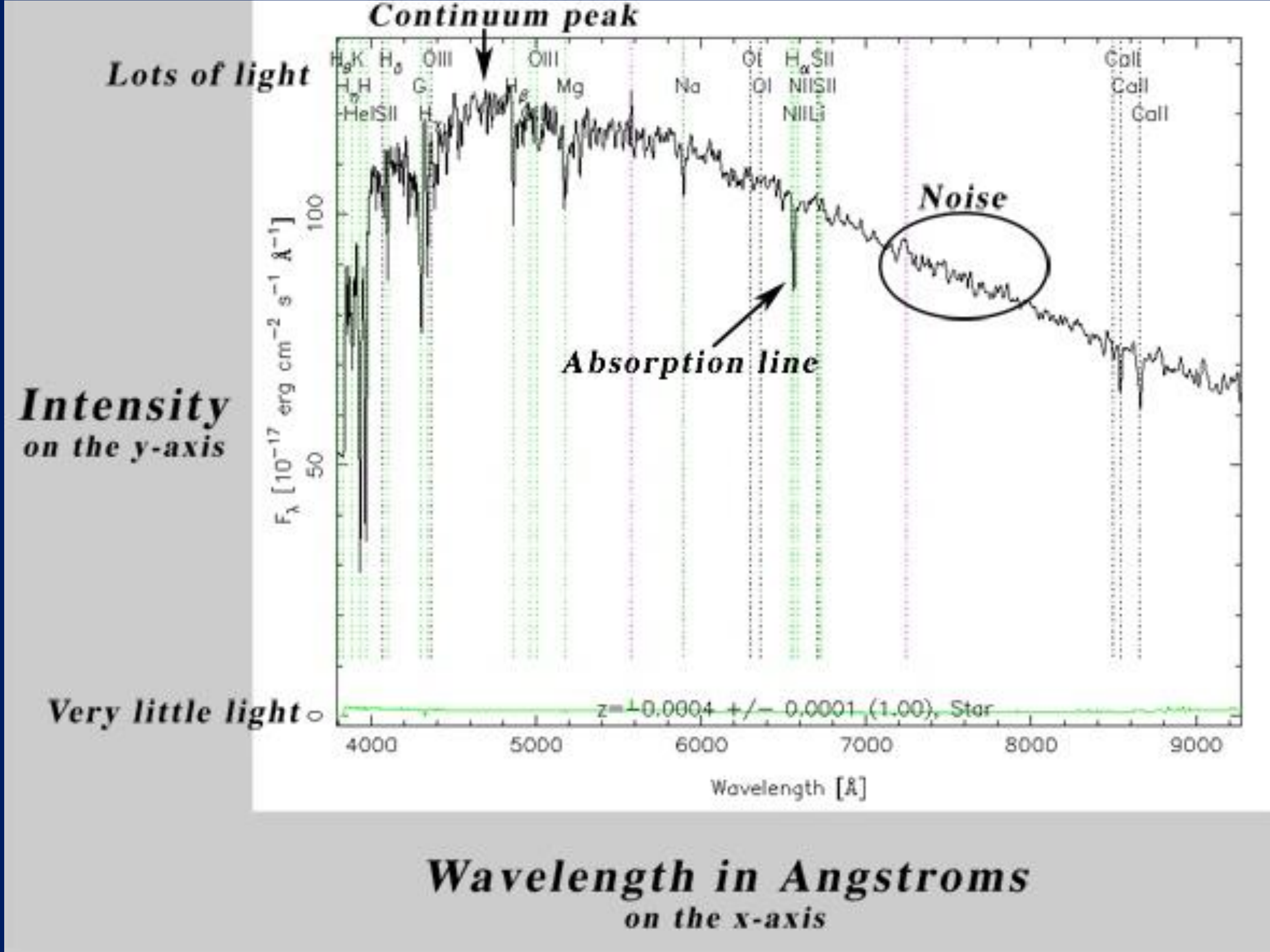
- Luminosity (brightness)
- Structure (sizes of layers; location of convective zones)
- Main sequence lifespan
- Temperature




Carroll & Ostlie (Pearson)

# Stellar Spectrum Example

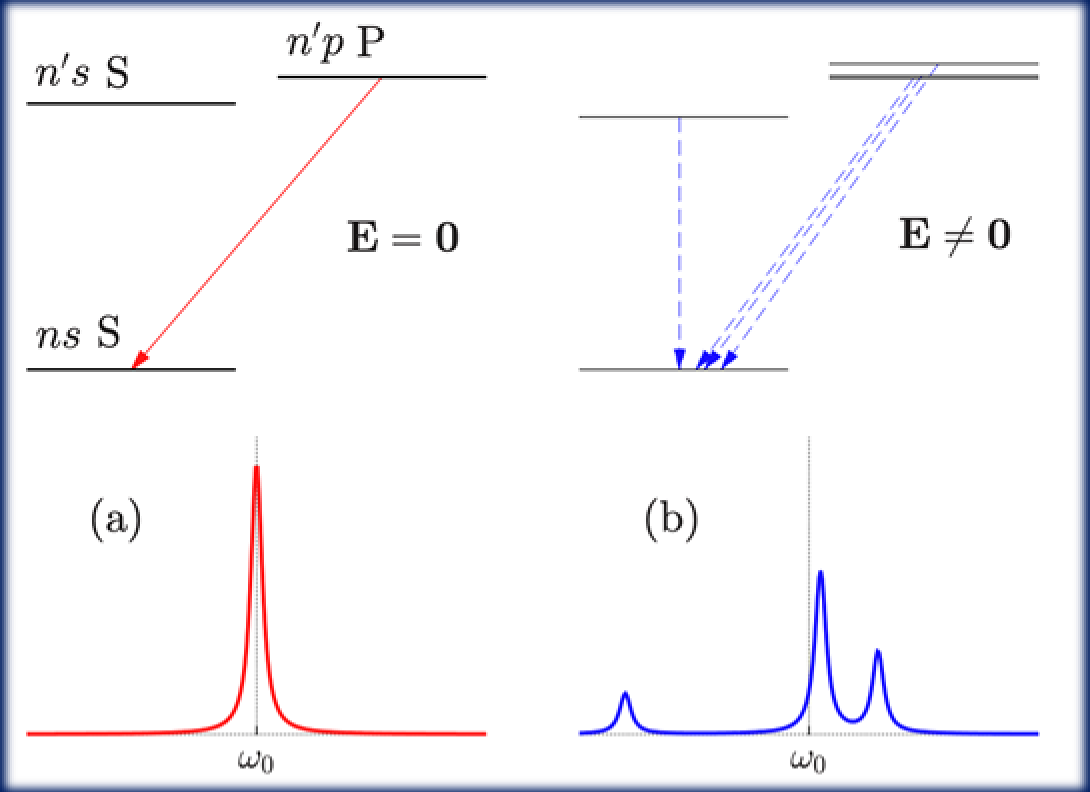



SDSS

# The presence of local electric fields (microfields) is known to affect line shapes

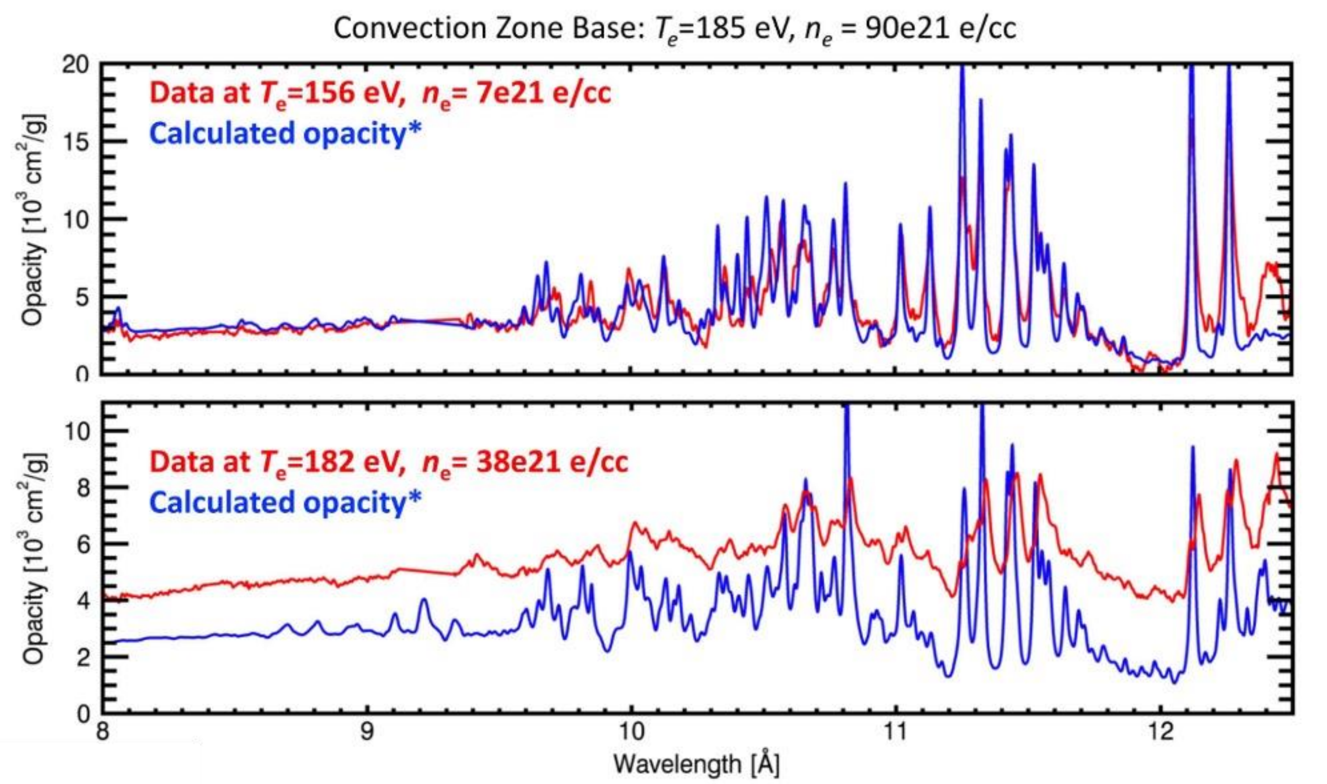
- Strongly coupled, dense plasmas have stochastic local electric fields
- Line broadening (Stark effect)
- Line merging
- Pressure ionization




Gigosos 2014

# We notice discrepancies between the observed and calculated line shapes of iron opacities

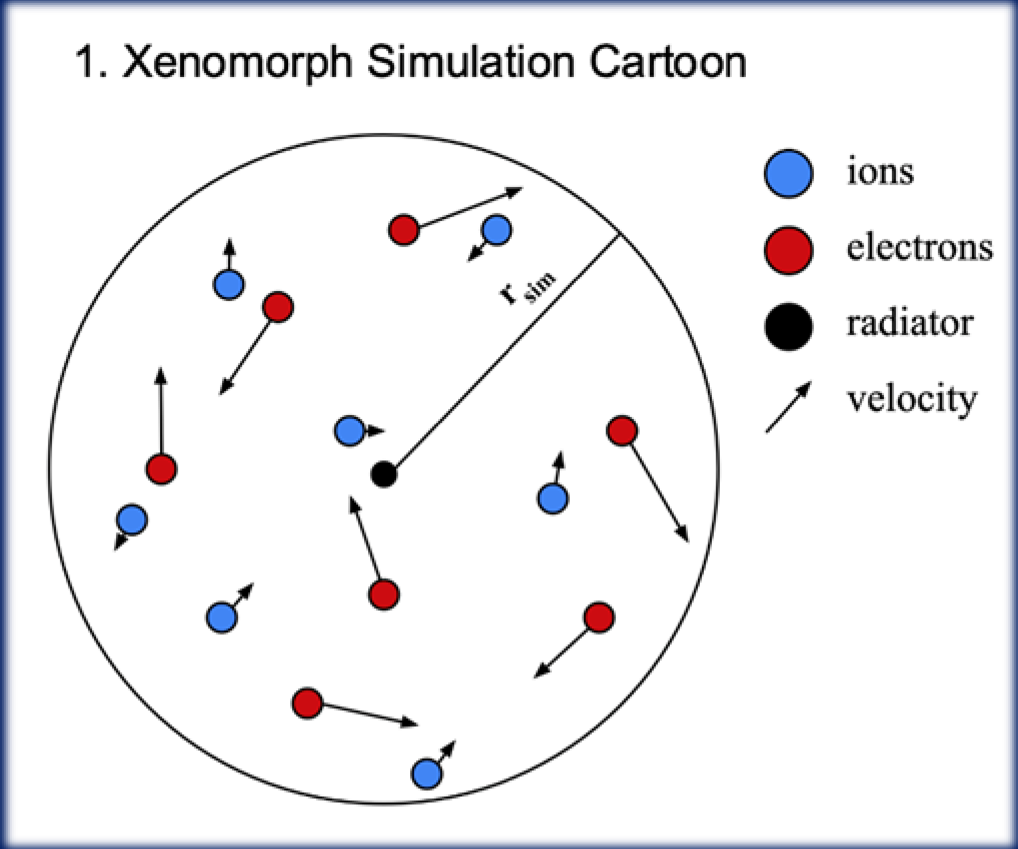
- Discrepancies at higher temperatures & densities
- Bound-bound (horizontal line) and bound-free (spectral lines) do not match




Bailey et al. 2015

# Existing opacity models, such as the line shape code Xenomorph, rely on semi-empirical methods

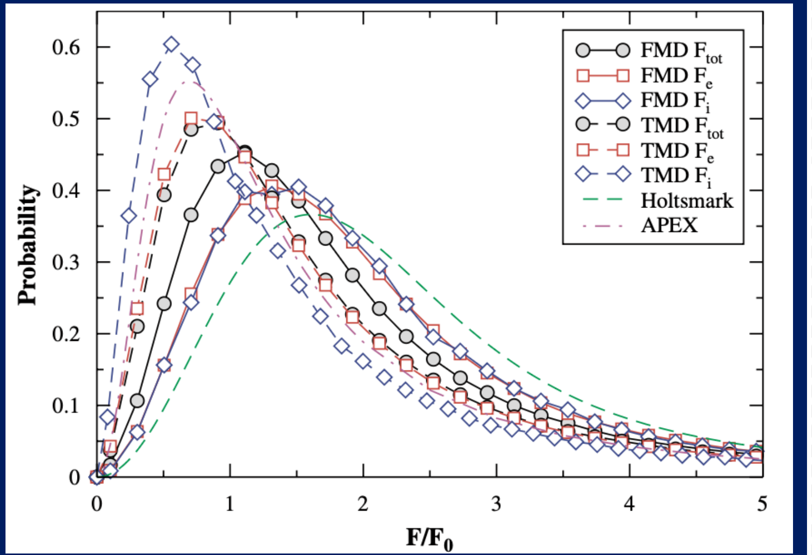
- Simulates straight-line trajectories
- Calculates microfields & uses them to create a time-dependent Hamiltonian
- Evolves with small timesteps
- Uses Fourier transform to calculate line shape



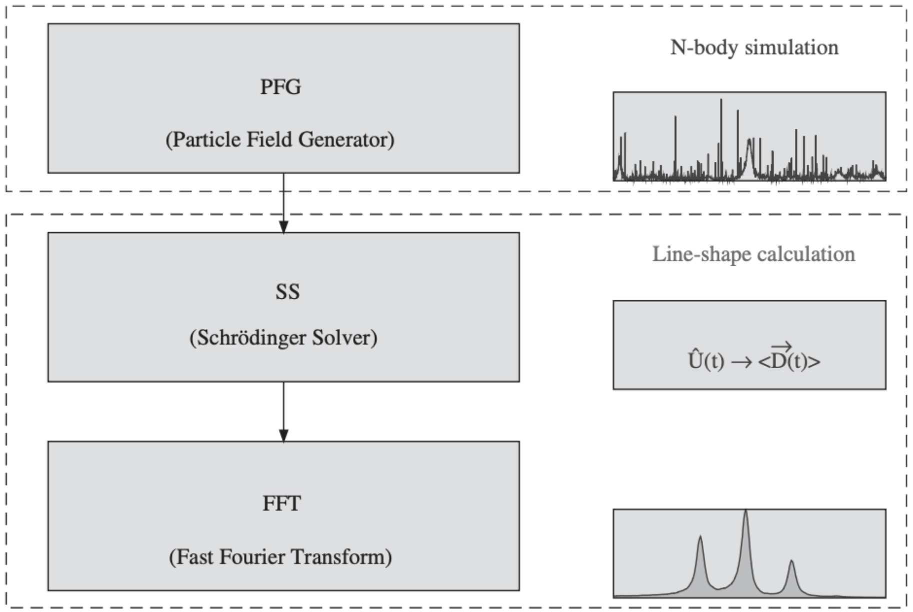

Jackson White

# Physically-motivated microfield generators may alleviate these spectral discrepancies

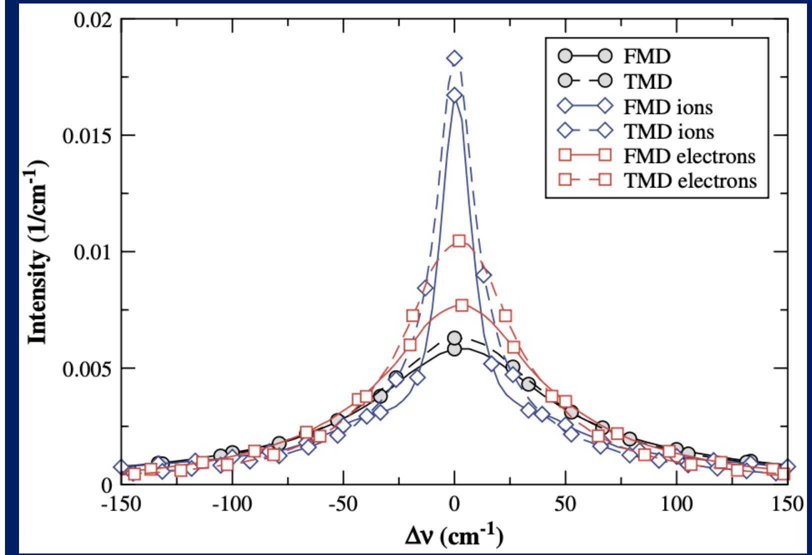
- Previous work has shown full molecular dynamics (MD) simulations affect microfield distributions and line shapes




Stambulchik et al. 2007



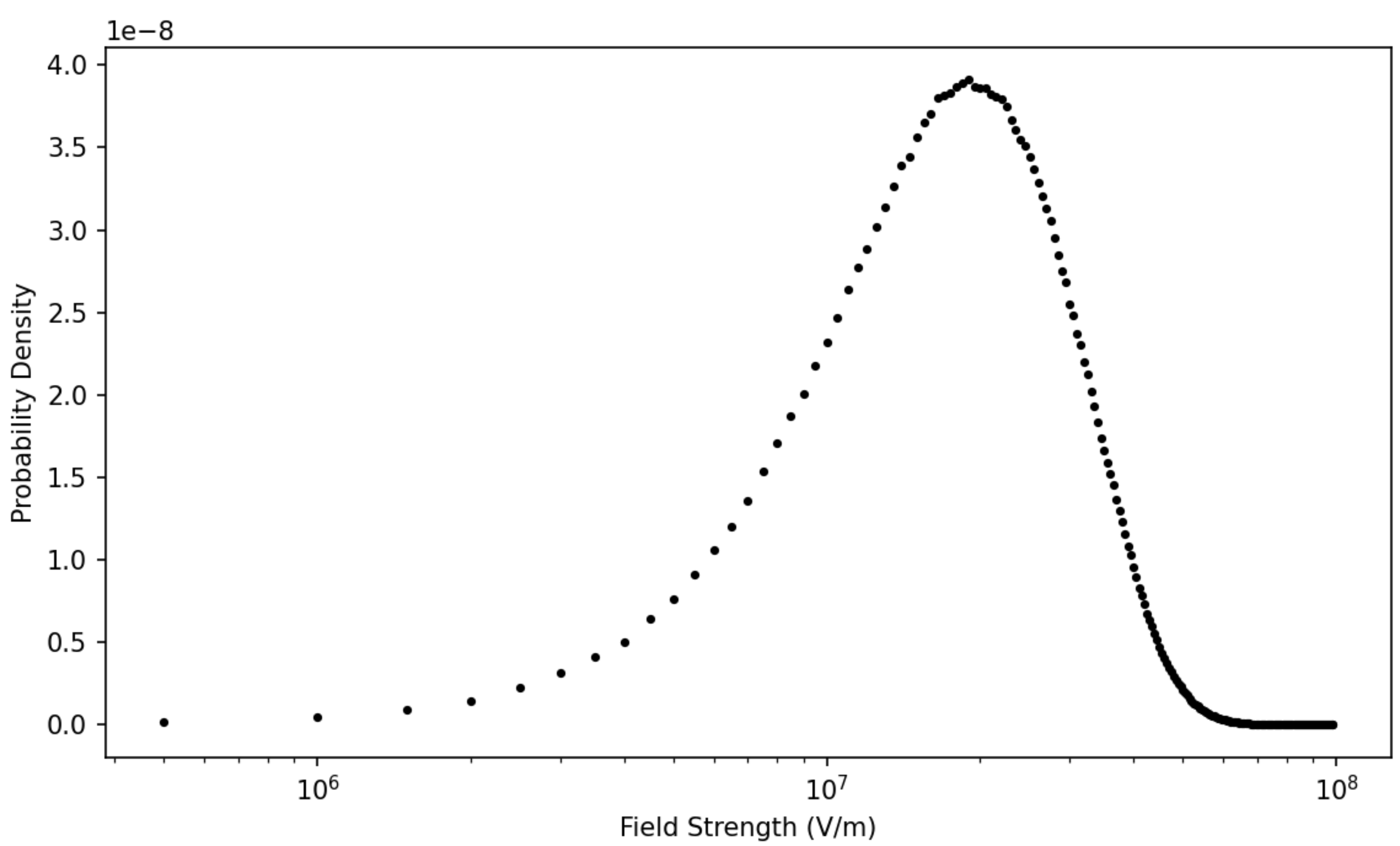

Stambulchik & Maron 2006




Stambulchik et al. 2007

# Preliminary microfield results and parameters for optimization study this academic year

- Real vs Fourier space computation ratio
- Size of Fourier space
- Simulation box size
- Time step
- Total simulation time
- Collision tolerance



# Thank you for your attention.

## References: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